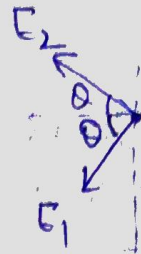
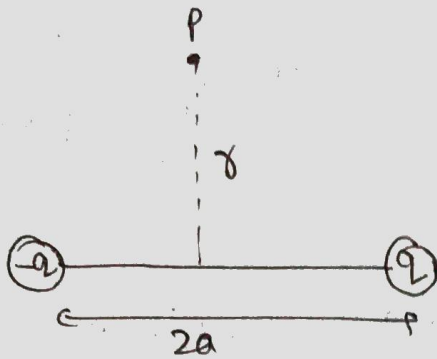


② at Equatorial Point



$$E_{\text{net}} = \frac{kP}{(r^2 + a^2)^{3/2}}$$

if $a \ll r$

$$E_{\text{net}} = \frac{kP}{r^3}$$

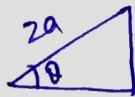
Torque due to a Dipole

As we know

$$\tau = F \times \perp \text{ Distance}$$

force exerted on charges $-q$ & $+q$

$$F = qE$$



$$\perp \text{ Dis.} = 2a \sin \theta$$

$$\tau = qE \times 2a \sin \theta$$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$\tau = pE \sin \theta$$

* if $\theta = 0^\circ$

$$\tau = 0 \text{ Stable}$$

if $\theta = 180^\circ$

$$\tau = 0$$

{ unstable }

if $\theta = 90^\circ$

$$\tau_{\text{max}} = pE$$